Lab Assignment 19.2

Program	**: B. Tech (CSE)**
Course Title	**: AI Assisted Coding**
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Task 1:

Prompt: Convert this python file handling code into java. Use proper java file streams and exception handling.

Python Code:

```python
file = open("example.txt", "w")
file.write("Hello, this is a sample text file.\n")
file.write("This file is used for demonstrating file handling operations in Python.\n")
file.close()
file = open("example.txt", "r")
content = file.read()
print("Contents of the file:")
print(content)
file.close()
file = open("example.txt", "a")
file.write("This line is added in append mode.\n")
file.close()
file = open("example.txt", "r")
updated_content = file.read()
print("Updated contents of the file:")
print(updated_content)
file.close()
file = open("example.txt", "w")
file.write("This file has been overwritten with new content.\n")
file.close()
file = open("example.txt", "r")
overwritten_content = file.read()
print("Overwritten contents of the file:")
print(overwritten_content)
file.close()
import os
os.remove("example.txt")
```

Java Code:

```java
import java.io.BufferedReader;
import java.io.BufferedWriter;
import java.io.File;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
public class Assign_19_2 {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        String filePath = "example.txt";
        // Write to the file
        try (BufferedWriter writer = new BufferedWriter(new FileWriter(filePath))) {
            writer.write(str: "Hello, this is a sample text file.\n");
            writer.write(str: "This file is used for demonstrating file handling operations in Java.\n");
        } catch (IOException e) {
            e.printStackTrace();
        }
        // Read from the file
        try (BufferedReader reader = new BufferedReader(new FileReader(filePath))) {
            String line;
            System.out.println(x: "Contents of the file:");
            while ((line = reader.readLine()) != null) {
                System.out.println(line);
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
        // Append to the file
        try (BufferedWriter writer = new BufferedWriter(new FileWriter(filePath, append: true))) {
            writer.write(str: "This line is added in append mode.\n");
        } catch (IOException e) {
            e.printStackTrace();
        }
        // Read the updated contents of the file
        try (BufferedReader reader = new BufferedReader(new FileReader(filePath))) {
            String line;
            System.out.println(x: "Updated contents of the file:");
            while ((line = reader.readLine()) != null) {
                System.out.println(line);
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
        // Overwrite the file
        try (BufferedWriter writer = new BufferedWriter(new FileWriter(filePath))) {
            writer.write(str: "This file has been overwritten with new content.\n");
        } catch (IOException e) {
            e.printStackTrace();
        }
        // Read the overwritten contents of the file
        try (BufferedReader reader = new BufferedReader(new FileReader(filePath))) {
            String line;
            System.out.println(x: "Overwritten contents of the file:");
            while ((line = reader.readLine()) != null) {
                System.out.println(line);
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
        // Delete the file
        File file = new File(filePath);
        if (file.delete()) {
            System.out.println(x: "File deleted successfully.");
        } else {
            System.out.println(x: "Failed to delete the file.");
        }
    }
}
```

Python code Output:

```
Contents of the file:
Hello, this is a sample text file.
This file is used for demonstrating file handling operations in Python.

Updated contents of the file:
Hello, this is a sample text file.
This file is used for demonstrating file handling operations in Python.
This line is added in append mode.

Overwritten contents of the file:
This file has been overwritten with new content.
```

C++ Code Output:

```
Contents of the file:
Hello, this is a sample text file.
This file is used for demonstrating file handling operations in Java.
Updated contents of the file:
Hello, this is a sample text file.
This file is used for demonstrating file handling operations in Java.
This line is added in append mode.
Overwritten contents of the file:
This file has been overwritten with new content.
File deleted successfully.
```

Explanation:

The Python program performs file handling operations such as writing, reading, appending, and deleting a file. It was successfully converted into a Java program using classes like BufferedWriter and BufferedReader for file operations. Exception handling is implemented using try-catch blocks to ensure safe execution. The Java program follows the same logic as the Python version and produces identical output.

Task 2:

Prompt: Convert this C++ sorting program into Python. Maintain same logic and output and add simple comments.

C++ Code:

```cpp
#include <iostream>
using namespace std;
// Function to perform selection sort
void selectionSort(int arr[], int n) {
    int i, j, min_idx;
    // One by one move boundary of unsorted subarray
    for (i = 0; i < n-1; i++) {
        // Find the minimum element in unsorted array
        min_idx = i;
        for (j = i+1; j < n; j++)
            if (arr[j] < arr[min_idx])
                min_idx = j;
        // Swap the found minimum element with the first element
        swap(arr[min_idx], arr[i]);
    }
}
// Function to print an array
void printArray(int arr[], int size) {
    int i;
    for (i = 0; i < size; i++)
        cout << arr[i] << " ";
    cout << endl;
}
// Driver code
int main() {
    int arr[] = {64, 25, 12, 22, 11};
    int n = sizeof(arr)/sizeof(arr[0]);
    selectionSort(arr, n);
    cout << "Sorted array: \n";
    printArray(arr, n);
    return 0;
}
```

Python Code:

```python
def selection_sort(arr):
    n = len(arr)
    # One by one move boundary of unsorted subarray
    for i in range(n):
        # Find the minimum element in unsorted array
        min_idx = i
        for j in range(i+1, n):
            if arr[j] < arr[min_idx]:
                min_idx = j
        # Swap the found minimum element with the first element
        arr[i], arr[min_idx] = arr[min_idx], arr[i]
# Function to print an array
def print_array(arr):
    for i in arr:
        print(i, end=' ')
    print()
# Driver code
if __name__ == "__main__":
    arr = [64, 25, 12, 22, 11]
    selection_sort(arr)
    print("Sorted array: ")
    print_array(arr)
```

C++ Code Output:

```
Sorted array:
11 12 22 25 64
```

Python Code Output:

```
Sorted array:
11 12 22 25 64
```

Explanation:

The C++ program implementing Selection Sort is successfully converted into Python. The logic remains the same, where the smallest element is selected from the unsorted portion and swapped with the current position. Python simplifies the syntax using direct swapping and dynamic arrays (lists). The program correctly sorts the array and prints the output in ascending order. Thus, both implementations produce the same result with equivalent functionality.

Task 3:

Prompt: Convert this SQL query with GROUP BY and aggregate functions into equivalent Pandas code using groupby().

SQL Code:

```sql
CREATE TABLE employees (
    id INTEGER,
    name TEXT,
    department TEXT,
    salary REAL
);

INSERT INTO employees VALUES
(1, 'A', 'HR', 30000),
(2, 'B', 'IT', 50000),
(3, 'C', 'HR', 35000),
(4, 'D', 'IT', 60000),
(5, 'E', 'Sales', 40000);

SELECT department, COUNT(*) AS employee_count, AVG(salary) AS average_salary
FROM employees
GROUP BY department;
```

Python Code:

```python
import pandas as pd
# Create a DataFrame to represent the employees table
data = {
    'id': [1, 2, 3, 4, 5],
    'name': ['A', 'B', 'C', 'D', 'E'],
    'department': ['HR', 'IT', 'HR', 'IT', 'Sales'],
    'salary': [30000, 50000, 35000, 60000, 40000]
}
employees_df = pd.DataFrame(data)
# Group by department and calculate employee count and average salary
result = employees_df.groupby('department').agg(employee_count=('id', 'count'), average_salary=('salary', 'mean')).reset_index()
print(result)
```

SQL Output:

```
HR|2|32500.0
IT|2|55000.0
Sales|1|40000.0
```

Python Code Output:

```
  department  employee_count  average_salary
0         HR               2         32500.0
1         IT               2         55000.0
2      Sales               1         40000.0
```

Explanation:
The given SQL query is converted into Pandas using a DataFrame to represent the employees table. The groupby('department') function is used to group data similar to SQL's GROUP BY clause. Aggregate functions like count and mean are applied using agg() to calculate employee count and average salary. The result is reset into a proper table format using reset_index(). The Pandas output matches the SQL query result, ensuring correct conversion.

Task 4:

Prompt: Convert this Python program into an equivalent C++ program. Maintain the same functionality and output.

Python Code:

```python
import pandas as pd
# Sample data
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Eve'],
    'Age': [25, 30, 35, 40, 45],
    'City': ['New York', 'Los Angeles', 'Chicago', 'Houston', 'Phoenix']
}
# Create a DataFrame
df = pd.DataFrame(data)
# Display the original DataFrame
print("Original DataFrame:")
print(df)
# Data processing: Filter out rows where Age is greater than 30
filtered_df = df[df['Age'] > 30]
print("\nFiltered DataFrame (Age > 30):")
print(filtered_df)
```

C++ code:

```cpp
#include <iostream>
#include <vector>
#include <string>
using namespace std;
// Structure to hold data
struct Person {
    string name;
    int age;
    string city;
};
int main() {
    // Sample data
    vector<Person> people = {
        {"Alice", 25, "New York"},
        {"Bob", 30, "Los Angeles"},
        {"Charlie", 35, "Chicago"},
        {"David", 40, "Houston"},
        {"Eve", 45, "Phoenix"}
    };
    // Display the original data
    cout << "Original Data:" << endl;
    for (const auto& person : people) {
        cout << person.name << ", " << person.age << ", " << person.city << endl;
    }
    // Data processing: Filter out rows where Age is greater than 30
    cout << "\nFiltered Data (Age > 30):" << endl;
    for (const auto& person : people) {
        if (person.age > 30) {
            cout << person.name << ", " << person.age << ", " << person.city << endl;
        }
    }
    return 0;
}
```

Python Code Output:

```
Original DataFrame:
      Name  Age         City
0    Alice   25     New York
1      Bob   30  Los Angeles
2  Charlie   35      Chicago
3    David   40      Houston
4      Eve   45      Phoenix

Filtered DataFrame (Age > 30):
      Name  Age     City
2  Charlie   35  Chicago
3    David   40  Houston
4      Eve   45  Phoenix
```

C++ Code Output:

```
Original Data:
Alice, 25, New York
Bob, 30, Los Angeles
Charlie, 35, Chicago
David, 40, Houston
Eve, 45, Phoenix

Filtered Data (Age > 30):
Charlie, 35, Chicago
David, 40, Houston
Eve, 45, Phoenix
```

Explanation:

Python code uses a DataFrame to store tabular data and filter rows, while the C++ version uses a struct Person and a vector to hold the same data. Printing all rows is done with a loop instead of print(df), and filtering (Age > 30) is handled with an if statement inside the loop. Essentially, C++ requires manual structuring and iteration, whereas Python handles it automatically with pandas.

Task 5:

Prompt: Convert this Python even/odd function into a C++ program.

Python Code:

```python
def check_even_odd(num):
    if num % 2 == 0:
        return "Even"
    else:
        return "Odd"
# Test cases
test_cases = [10, 15, 0]
for test in test_cases:
    result = check_even_odd(test)
    print(f"{test} is {result}.")
```

C++ Code:

```cpp
#include <iostream>
#include <vector>
#include <string>
std::string check_even_odd(int num) {
    if (num % 2 == 0) {
        return "Even";
    } else {
        return "Odd";
    }
}
int main() {
    std::vector<int> test_cases = {10, 15, 0};
    for (int test : test_cases) {
        std::string result = check_even_odd(test);
        std::cout << test << " is " << result << "." << std::endl;
    }
    return 0;
}
```

Python Code Output:

```
10 is Even.
15 is Odd.
0 is Even.
```

C++ Code Output:

```
10 is Even.
15 is Odd.
0 is Even.
```

Explanation:

- The function check_even_odd takes an integer and returns "Even" or "Odd" based on % 2.
- The test_cases vector stores numbers to check, like the Python list.
- A range-based for loop iterates over each number in test_cases.
- std::cout prints the result in the same format as Python's print.
- The program mimics Python behaviour but uses C++ syntax and types (std::string, std::vector).